

AP-E3 Global-Definition and Deep-UV Audit: Differential Characters on Non-Extendible Four-Sectors and the Exact Boundary of the $SU(3)$ Embedding

Route F independent research note

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Abstract

This note closes the ordinary global-definition problem for the level-two mixed Wess–Zumino–Witten functional and sharply delimits what remains UV data. For the actual two-flavour target $X = SU(2) \times S^2 \simeq S^3 \times S^2$, the integral curvature $n\omega_3 \wedge \omega_2$ has a unique degree-five differential-character refinement. Its holonomy defines the mixed action on every closed oriented four-manifold and every sigma-model map, including configurations that do not extend over a spin five-manifold. A Čech–Deligne representative and its descent equations make the construction local and computable.

Spin bordism contributes additional information that the curvature cannot fix. Stable splitting gives

$$\Omega_4^{\text{Spin}}(S^3 \times S^2) \simeq \mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2,$$

not merely one reduced $\mathbb{Z}_2$. The two reduced signs are detected by the spin one-manifold $g^{-1}(p_3)$ and the Arf invariant of the spin two-manifold $s^{-1}(p_2)$. The full action is therefore

$$Z_{n,\epsilon_3,\epsilon_2} = \text{Hol}_{\widehat{\Omega}_n}(g, s)(-1)^{\epsilon_3\nu_3 + \epsilon_2\nu_2}.$$

The two signs require a specified UV APS/Dai–Freed determinant and are not silently chosen here.

The proposed all-scale breaking $SU(3) \rightarrow [SU(2)_c \times U(1)_g]/\mathbb{Z}_2$ is then audited exactly. Its kernel forces $2j + q = 0 \pmod{2}$, so a colour singlet has even abelian charge and the original charge-one scalar cannot occur in any $SU(3)$ representation. Two $\mathbf{\bar{3}}$ scalars do furnish a conditional charge-two variant, and explicit adjoint mass matrices can make their coloured partners heavy. Vectorlike fundamental fermions give $\mathbf{3} \rightarrow \mathbf{2}_{+1} \oplus \mathbf{1}_{-2}$; all dynamical gauge anomalies cancel, and an adjoint mass matrix can tune the quarks light while making the extra leptons heavy. Neither tuning is symmetry protected. More decisively, the charge-two scalar needs only one conjugate dressing of qq , giving a doublet rather than the exactly-two triplet. Thus the global mixed-WZW functional is defined, while a Route-E-equivalent all-scale UV completion is not established.

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1 Scope, conventions, and claim discipline

The purpose of this note is narrow. It does not repeat the local anomaly ledger or the worldline reduction already established in the main AP-E3 note. Instead it addresses two residual questions:

- (Q1) how to define the mixed functional when the four-dimensional spin configuration is not a five-dimensional spin boundary;
- (Q2) whether the suggested $SU(3)$ embedding actually reproduces the charge-one scalar, decouples all partners, and preserves the level-two triplet mechanism.

We use

$$X = SU(2) \times S^2 \simeq S^3 \times S^2, \quad \int_{S^3} \omega_3 = 1, \quad \int_{S^2} \omega_2 = 1, \quad (1)$$

and the mixed curvature

$$\Omega_{5,n} = n \omega_3 \wedge \omega_2, \quad n = 2. \quad (2)$$

The trace in $\omega_3 = (24\pi^2)^{-1} \text{Tr}(g^{-1}dg)^3$ is fixed by Eq. (1), so no later sign or normalization is hidden in a representation convention.

Statements are marked as follows. **Proved** means a topological, group-theoretic, or algebraic result follows from the displayed assumptions. **Conditional** means an explicit Lagrangian or mass matrix exists but its hierarchy or radiative stability is assumed. **Open** means no completion is claimed.

2 The extension-free differential-character functional

2.1 Integral class and uniqueness

Let $u_3 \in H^3(S^3; \mathbb{Z})$ and $u_2 \in H^2(S^2; \mathbb{Z})$ be positive generators. Künneth’s theorem gives

$$H^\bullet(X; \mathbb{Z}) \simeq \Lambda_{\mathbb{Z}}(u_3) \otimes \frac{\mathbb{Z}[u_2]}{(u_2^2)}, \quad H^4(X; \mathbb{Z}) = 0, \quad H^5(X; \mathbb{Z}) = \mathbb{Z}\{u_3 u_2\}. \quad (3)$$

A degree- k Cheeger–Simons differential character is a homomorphism

$$\widehat{h} : Z_{k-1}(X) \longrightarrow \mathbb{R}/\mathbb{Z} \quad (4)$$

for which there is a closed integral-period k -form $\text{curv}(\widehat{h})$ satisfying

$$\widehat{h}(\partial C_k) = \int_{C_k} \text{curv}(\widehat{h}) \pmod{\mathbb{Z}}. \quad (5)$$

Differential characters are equivalent to smooth Deligne cohomology [1, 2].

Choose the unique normalized refinements

$$\widehat{u}_3 \in \widehat{H}^3(S^3; \mathbb{Z}), \quad \widehat{u}_2 \in \widehat{H}^2(S^2; \mathbb{Z}), \quad (6)$$

with curvatures ω_3 and ω_2 . Their differential cup product is

$$\widehat{\Omega}_n := n \widehat{u}_3 \smile \widehat{u}_2 \in \widehat{H}^5(X; \mathbb{Z}), \quad (7)$$

and obeys

$$\begin{aligned} \text{curv}(\widehat{\Omega}_n) &= n\omega_3 \wedge \omega_2, \\ c(\widehat{\Omega}_n) &= n u_3 u_2. \end{aligned} \quad (8)$$

Theorem 2.1 (Unique ordinary differential refinement). *For every $n \in \mathbb{Z}$, Eq. (7) is the unique ordinary degree-five differential character with curvature $n\omega_3 \wedge \omega_2$.*

Proof. The curvature exact sequence is

$$0 \longrightarrow H^4(X; \mathbb{R}/\mathbb{Z}) \longrightarrow \widehat{H}^5(X; \mathbb{Z}) \xrightarrow{\text{curv}} \Omega_{\mathbb{Z}}^5(X) \longrightarrow 0. \quad (9)$$

Equation (3) and the universal coefficient theorem give $H^4(X; \mathbb{R}/\mathbb{Z}) = 0$. Hence the fiber over the prescribed integral curvature contains exactly one character. $\square$

2.2 Definition on an arbitrary four-sector

Let M_4 be a closed oriented four-manifold and $\Phi = (g, s) : M_4 \rightarrow X$. Define

$$Z_n^{\text{diff}}[M_4, \Phi] := \exp \left[2\pi i \widehat{\Omega}_n(\Phi_*[M_4]) \right]. \quad (10)$$

This formula requires neither a five-manifold nor an extension of Φ . It is therefore a definition, rather than an instruction that fails in a non-bounding sector.

When a singular five-chain C_5 in X satisfies $\partial C_5 = \Phi_*[M_4]$, the character axiom gives

$$Z_n^{\text{diff}}[M_4, \Phi] = \exp \left[2\pi i n \int_{C_5} \omega_3 \wedge \omega_2 \right]. \quad (11)$$

Such a chain exists here because $H_4(X; \mathbb{Z}) = 0$. It need not be the image of a smooth spin five-manifold. If a genuine extension $(Y_5, \widetilde{\Phi})$ exists, Eq. (11) reduces to

$$Z_n^{\text{diff}} = \exp \left[2\pi i n \int_{Y_5} \widetilde{\Phi}^*(\omega_3 \wedge \omega_2) \right]. \quad (12)$$

Changing the extension adds an integral period and leaves the exponential unchanged. Thus the five-dimensional formula is a consequence of the character, not its universal definition.

3 A computable Čech–Deligne representative

3.1 Descent data and triangulated holonomy

On a good cover $\{U_i\}$ of X , a degree-five Deligne cocycle may be written

$$\check{C} = (C_i^{(4)}, C_{ij}^{(3)}, C_{ijk}^{(2)}, C_{ijkl}^{(1)}, C_{ijklm}^{(0)}, N_{ijklmn}), \quad (13)$$

where N is integer valued. We fix the descent-sign convention

$$\begin{aligned} dC_i^{(4)} &= \Omega_{5,n}, \\ \delta C^{(4)} &= dC^{(3)}, & \delta C^{(3)} &= -dC^{(2)}, \\ \delta C^{(2)} &= dC^{(1)}, & \delta C^{(1)} &= -dC^{(0)}, \\ \delta C^{(0)} &= N, & \delta N &= 0. \end{aligned} \quad (14)$$

The integer cocycle represents nu_3u_2 . Another representative differs by a Deligne coboundary and defines the same character.

Choose a triangulation subordinate to the cover and a barycentric labelling. The logarithm of the holonomy is the alternating flag sum

$$\begin{aligned} \frac{1}{2\pi i} \log \text{Hol}_{\check{C}}(M_4) &= \sum_{\sigma^4} \int_{\sigma^4} C_{i_0}^{(4)} - \sum_{\sigma^3 \subset \sigma^4} \int_{\sigma^3} C_{i_0 i_1}^{(3)} + \sum_{\sigma^2 \subset \sigma^3 \subset \sigma^4} \int_{\sigma^2} C_{i_0 i_1 i_2}^{(2)} \\ &\quad - \sum_{\sigma^1 \subset \dots \subset \sigma^4} \int_{\sigma^1} C_{i_0 i_1 i_2 i_3}^{(1)} + \sum_{\sigma^0 \subset \dots \subset \sigma^4} C_{i_0 i_1 i_2 i_3 i_4}^{(0)} \pmod{\mathbb{Z}}. \end{aligned} \quad (15)$$

The orientations and incidence numbers are those of the nested simplices. Equations (14) make internal faces cancel, while $N \in \mathbb{Z}$ makes cover changes and refinements invisible after exponentiation. This is the promised extension-free local algorithm.

3.2 Explicit normalization check

Use hyperspherical coordinates on S^3 :

$$\omega_3 = \frac{1}{2\pi^2} \sin^2 \chi \sin \theta \, d\chi \wedge d\theta \wedge d\varphi. \quad (16)$$

Two illustrative gerbe potentials are

$$\begin{aligned} B_N &= f_N(\chi) \sin \theta \, d\theta \wedge d\varphi, & f_N(\chi) &= \frac{2\chi - \sin 2\chi}{8\pi^2}, \\ B_S &= f_S(\chi) \sin \theta \, d\theta \wedge d\varphi, & f_S(\chi) &= f_N(\chi) - \frac{1}{4\pi}. \end{aligned} \quad (17)$$

They satisfy

$$dB_N = dB_S = \omega_3, \quad \int_{S^2} (B_N - B_S) = 1. \quad (18)$$

The two-chart presentation may be refined to a good cover; its overlap already exposes the integral gerbe transition.

For the S^2 factor, Route-E conventions give

$$\begin{aligned} A_N &= \frac{1 - \cos \Theta}{2} d\Phi, & A_S &= -\frac{1 + \cos \Theta}{2} d\Phi, \\ A_N - A_S &= d\Phi, & \omega_2 &= \frac{dA}{2\pi} = \frac{\sin \Theta}{4\pi} d\Theta \wedge d\Phi. \end{aligned} \quad (19)$$

Locally one may start with

$$C_{N,\alpha}^{(4)} = nB_N \wedge \omega_2, \quad C_{S,\alpha}^{(4)} = nB_S \wedge \omega_2. \quad (20)$$

The non-exact integral overlap in Eq. (18) is precisely why a single global four-form does not exist. The remaining terms in Eq. (13) are generated by the Deligne cup product of the gerbe and Hopf-line cocycles. Omitting them would make the action patch dependent.

4 Spin bordism, the two torsion signs, and APS selection

4.1 The special $N_f = 2$ stable splitting

The target in this repository uses $SU(2) \simeq S^3$, so its bordism must not be replaced by a generic $SU(N \geq 3)$ result. The stable splitting is

$$\Sigma^\infty(S^3 \times S^2)_+ \simeq S^0 \vee S^2 \vee S^3 \vee S^5. \quad (21)$$

Since spin bordism is a generalized homology theory and

$$\Omega_0^{\text{Spin}} = \mathbb{Z}, \quad \Omega_1^{\text{Spin}} = \mathbb{Z}_2, \quad \Omega_2^{\text{Spin}} = \mathbb{Z}_2, \quad \Omega_3^{\text{Spin}} = 0, \quad \Omega_4^{\text{Spin}} = \mathbb{Z}, \quad (22)$$

we obtain the following result.

Theorem 4.1 (Full and reduced four-bordism).

$$\begin{aligned} \Omega_4^{\text{Spin}}(S^3 \times S^2) &\simeq \mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2, \\ \tilde{\Omega}_4^{\text{Spin}}(S^3 \times S^2) &\simeq \mathbb{Z}_2 \oplus \mathbb{Z}_2. \end{aligned} \quad (23)$$

The free $\mathbb{Z}$ is the field-independent gravitational sector. The first reduced $\mathbb{Z}_2$ comes from the S^3 factor and the second from S^2 .

The S^3 summand is the special two-flavour discrete WZW sector discussed in Ref. [3]. The extra S^2 summand is the one retained in the generic mixed-target calculation of Ref. [4]. Both are present for the actual product $S^3 \times S^2$.

4.2 Transverse inverse-image invariants

Let $p_3 \in S^3$ and $p_2 \in S^2$ be regular values. Transversality gives

$$\begin{aligned} L_g &:= g^{-1}(p_3), & \dim L_g &= 1, \\ \Sigma_s &:= s^{-1}(p_2), & \dim \Sigma_s &= 2. \end{aligned} \quad (24)$$

The oriented tangent spaces at the regular values frame the normal bundles. Together with the spin structure on M_4 , these framings induce spin structures on L_g and Σ_s . Define

$$\nu_3(M_4, g) = [L_g] \in \Omega_1^{\text{Spin}} \simeq \mathbb{Z}_2, \quad (25)$$

$$\nu_2(M_4, s) = \text{Arf}(\Sigma_s) \in \Omega_2^{\text{Spin}} \simeq \mathbb{Z}_2. \quad (26)$$

Changing the regular value gives a spin bordism between inverse images, so the invariants are well defined. Representatives are

$$\begin{aligned} (S_{\text{periodic}}^1 \times S^3, g = \text{pr}_{S^3}, s = \text{const}) & \quad \text{for } \nu_3 = 1, \\ (T_{\text{odd}}^2 \times S^2, g = \text{const}, s = \text{pr}_{S^2}) & \quad \text{for } \nu_2 = 1. \end{aligned} \quad (27)$$

Definition 4.2 (Full spin-refined mixed functional). After factoring out the pure gravitational phase, the most general spin-refined action with curvature Eq. (2) is

$$Z_{n,\epsilon_3,\epsilon_2}[M_4, g, s] = Z_n^{\text{diff}}[M_4, g, s] (-1)^{\epsilon_3 \nu_3(M_4, g) + \epsilon_2 \nu_2(M_4, s)}, \quad \epsilon_3, \epsilon_2 \in \mathbb{Z}_2. \quad (28)$$

The ordinary differential character fixes the continuous/integral mixed part uniquely. It does not fix either torsion character. On a spin-bounding configuration both ν_3 and ν_2 vanish, so all four choices reduce to Eq. (12); this explains why extension tests alone cannot distinguish them.

4.3 What APS/Dai–Freed must compute

The coefficients (ϵ_3, ϵ_2) are selected by the phase of a specified UV determinant, not by choosing a convenient square root. For a real or pseudoreal one-parameter family of regulated boundary operators $\mathcal{D}_t$, the mod-two spectral flow satisfies

$$\text{SF}_2(\mathcal{D}_t) = \text{ind}_{2,\text{APS}}(\partial_t + \mathcal{D}_t). \quad (29)$$

For a complex family, the corresponding object is the exponentiated reduced eta invariant

$$\tau_Y(\mathcal{D}) = \exp[2\pi i \bar{\eta}(\mathcal{D}_Y)], \quad \bar{\eta} = \frac{\eta + \dim \ker \mathcal{D}}{2}, \quad (30)$$

viewed as a section of the determinant line. APS gluing makes the result a bordism character [5, 6].

To determine Eq. (28), one must evaluate the UV determinant on both backgrounds in Eq. (27). The standard two-flavour QCD result gives $\epsilon_3 = n_c \bmod 2 = 0$ for two colours, provided the usual quark determinant and skyrmion identification are retained. No calculation in the present repository fixes ϵ_2 . A convention $\epsilon_2 = 0$ is allowed but is not a first-principles derivation. Modern bordism treatments illustrate precisely why this determinant step is independent of local anomaly matching [7, 8].

Fail-closed gate 4.3 (Global-definition result). The ordinary non-extendible mixed functional is now defined by Eq. (10). The full spin theory remains a two-bit family until a UV APS/Dai–Freed calculation selects (ϵ_3, ϵ_2) . Calling the entire spin refinement *closed* before that calculation would be false.

5 The exact $SU(3)$ quotient and branching

5.1 Group homomorphism and parity theorem

Take an adjoint field $\Sigma \sim \mathbf{8}$ with

$$\langle \Sigma \rangle = V_\Sigma T, \quad T = \text{diag}(1, 1, -2). \quad (31)$$

The unbroken subgroup is the image of

$$\iota : SU(2)_c \times U(1)_g \longrightarrow SU(3), \quad \iota(U, e^{i\alpha}) = \begin{pmatrix} e^{i\alpha} U & 0 \\ 0 & e^{-2i\alpha} \end{pmatrix}. \quad (32)$$

Its kernel is

$$\ker \iota = \{(\mathbf{1}, 1), (-\mathbf{1}, -1)\} \simeq \mathbb{Z}_2, \quad (33)$$

so the gauge group below V_Σ is

$$U(2) \simeq [SU(2)_c \times U(1)_g] / \mathbb{Z}_2. \quad (34)$$

Theorem 5.1 (Quotient-parity rule). *An $SU(2)_c$ spin- j multiplet of integer-normalized abelian charge q is a representation of Eq. (34) if and only if*

$$2j + q = 0 \pmod{2}. \quad (35)$$

Consequently every colour doublet has odd charge and every colour singlet has even charge.

Proof. The kernel generator acts as $(-1)^{2j} e^{i\pi q}$. It must act trivially, which is exactly Eq. (35). $\square$

This is a representation-independent obstruction. In particular, a $\mathbf{1}_{+1}$ scalar cannot descend from any $SU(3)$ representation. It is not repaired by placing that scalar next to coloured partners; the forbidden singlet itself never appears.

5.2 Low-representation decomposition

With the normalization in Eq. (31), direct weight decomposition gives [9]

$SU(3)$ irrep	$SU(2)_c \times U(1)_g$ content
3	$\mathbf{2}_{+1} \oplus \mathbf{1}_{-2}$
$\bar{\mathbf{3}}$	$\mathbf{2}_{-1} \oplus \mathbf{1}_{+2}$
6	$\mathbf{3}_{+2} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_{-4}$
$\bar{\mathbf{6}}$	$\mathbf{3}_{-2} \oplus \mathbf{2}_{+1} \oplus \mathbf{1}_{+4}$
8	$\mathbf{3}_0 \oplus \mathbf{2}_{+3} \oplus \mathbf{2}_{-3} \oplus \mathbf{1}_0$
10	$\mathbf{4}_{+3} \oplus \mathbf{3}_0 \oplus \mathbf{2}_{-3} \oplus \mathbf{1}_{-6}$

Every term obeys Eq. (35). The companion verifier derives the symmetric-power rows from weights rather than storing only their dimensions.

6 Fermions, anomaly matching, and lepton decoupling

6.1 Vectorlike deep-UV multiplets

Take $N_f = 2$ Dirac fields $\Psi_i \in \mathbf{3}$ of $SU(3)$. Below V_Σ ,

$$\Psi_i \longrightarrow q_i \oplus \ell_i, \quad q_i \sim \mathbf{2}_{+1}, \quad \ell_i \sim \mathbf{1}_{-2}. \quad (36)$$

In all-left notation the fields are

field	$SU(2)_c$	$U(1)_g$
q_L	2	+1
q_R^c	2	-1
ℓ_L	1	-2
ℓ_R^c	1	+2

The extra lepton is not optional: it is the third component of every fundamental $SU(3)$ fermion.

The deep theory is vectorlike, hence

$$\mathcal{A}_{SU(3)^3} = N_f[A(\mathbf{3}) + A(\bar{\mathbf{3}})] = 0. \quad (37)$$

After branching, the complete local ledger is

$$\begin{aligned} \mathcal{A}_{SU(2)_c^2 U(1)_g} &= N_f T(\mathbf{2})(1 - 1) = 0, \\ \mathcal{A}_{U(1)_g^3} &= N_f \{2(1^3 - 1^3) + (-2)^3 + 2^3\} = 0, \\ \mathcal{A}_{\text{grav}^2 U(1)_g} &= N_f \{2(1 - 1) - 2 + 2\} = 0. \end{aligned} \quad (38)$$

There are $2N_f = 4$ left-Weyl colour doublets, so the colour Witten anomaly also vanishes.

For independent quark and lepton chiral-flavour backgrounds, the operator-valued mixed coefficients are

$$\widehat{\kappa}_q = 2(+1) = +2, \quad \widehat{\kappa}_\ell = 1(-2) = -2. \quad (39)$$

They cancel on the diagonal deep-UV flavour group, as required because the simple $SU(3)$ phase has no exact magnetic $U(1)$ one-form symmetry [4]. For $N_f = 2$, perturbative flavour cubic anomalies vanish, but the global Witten count is informative: the quark flavour factor sees two copies (even), the lepton factor one copy (odd), and the diagonal deep factor three copies (odd). Decoupling the lepton while keeping an exact unchanged deep chiral symmetry is therefore not innocuous.

6.2 Adjoint mass matrix and its exact conditions

The most direct renormalizable splitting is

$$\mathcal{L}_{\text{mass}} = -\bar{\Psi}_{Li}(m_{ij} + y_{ij}\Sigma)\Psi_R^j + \text{h.c.}, \quad (40)$$

where m and y are complex $N_f \times N_f$ matrices. The two blocks are

$$M_q = m + V_\Sigma y, \quad M_\ell = m - 2V_\Sigma y. \quad (41)$$

Proposition 6.1 (Tree-level decoupling condition). *All two quark flavours are massless and all two leptons are massive at tree level if*

$$m = -V_\Sigma y, \quad \det y \neq 0. \quad (42)$$

Then

$$M_q = 0, \quad M_\ell = -3V_\Sigma y, \quad \sigma_{\min}(M_\ell) = 3V_\Sigma \sigma_{\min}(y) > 0. \quad (43)$$

This proves a mass-matrix identity, not natural decoupling. Both terms in Eq. (40) connect left and right flavour groups. If the exact deep symmetry is $SU(N_f)_L \times SU(N_f)_R$ and Σ is a flavour singlet, they are forbidden. If they are present, only vector flavour is exact and the quark chiral group is accidental below the thresholds. In the absence of a protecting symmetry, the cancellation $M_q = 0$ must be retuned after heavy gauge, adjoint, and lepton loops; parametrically,

$$\delta M_q \sim \frac{g_3^2}{16\pi^2} y V_\Sigma \quad (44)$$

up to representation- and scheme-dependent coefficients. The emergence of quark flavour, rather than its exact deep-UV realization, is required by the generalized-symmetry no-go theorem.

Fail-closed gate 6.2 (Fermion decoupling). Equations (42)–(43) are [proved tree-level algebra](#). A symmetry-protected or fully retuned quantum decoupling with the required emergent chiral theory is [open](#). Adding spectators changes both the flavour-Witten and determinant-phase ledgers and must be audited as a new model.

7 Scalar embedding: the no-go and the charge-two variant

7.1 The charge-one model cannot be embedded

The intermediate AP-E3 model used a colour-singlet scalar doublet $\phi_I \sim \mathbf{1}_{+1}$, $I = 1, 2$. The quotient-parity theorem proves that this representation does not exist in the $SU(3)$ embedding. There is a second, directly physical form of the obstruction.

Let the quark doublet have positive charge X_q and a colour-singlet scalar have positive charge X_ϕ . Equation (35) implies

$$X_q \in 2\mathbb{Z} + 1, \quad X_\phi \in 2\mathbb{Z}. \quad (45)$$

A two-quark baryon dressed by m conjugate scalars is neutral only if

$$2X_q - mX_\phi = 0. \quad (46)$$

For exactly two dressings, $m = 2$ requires $X_q = X_\phi$, contradicting Eq. (45).

Theorem 7.1 ($SU(3)$ exactly-two no-go). *No representation of $[SU(2)_c \times U(1)_g]/\mathbb{Z}_2$ descending from $SU(3)$ can simultaneously contain an odd-charge colour-doublet quark, an equal-charge colour-singlet scalar, and the neutral local operator $qq(\phi^\dagger)^2$. The AP-E3 exactly-two triplet mechanism therefore does not lift unchanged to this $SU(3)$ embedding.*

7.2 A concrete charge-two scalar branch

There is nevertheless a nearby, anomaly-consistent scalar spectrum. Take two complex fields

$$\Phi_I \in \bar{\mathbf{3}}, \quad I = 1, 2, \quad (47)$$

forming a global $SU(2)_\phi$ doublet. They decompose as

$$\Phi_I \longrightarrow \chi_I \oplus \phi_I, \quad \chi_I \sim \mathbf{2}_{-1}, \quad \phi_I \sim \mathbf{1}_{+2}. \quad (48)$$

The coloured fields χ_I are the unavoidable partners.

The quadratic potential may be written

$$V_2 = \Phi_I^\dagger [(m^2)_{IJ} + \kappa_{IJ} \Sigma_{\bar{3}} + \rho_{IJ} \Sigma_{\bar{3}}^2] \Phi_J, \quad \Sigma_{\bar{3}} = -\Sigma_3^T. \quad (49)$$

After Eq. (31),

$$\begin{aligned} M_\chi^2 &= m^2 - V_\Sigma \kappa + V_\Sigma^2 \rho, \\ M_\phi^2 &= m^2 + 2V_\Sigma \kappa + 4V_\Sigma^2 \rho. \end{aligned} \quad (50)$$

Exact $SU(2)_\phi$ requires each coefficient matrix to be proportional to δ_{IJ} . A sufficient tree-level window is

$$\lambda_{\text{eff}} > 0, \quad M_\phi^2 = -\mu_\phi^2 < 0, \quad M_\chi^2 + \lambda_{\chi\phi} v_\phi^2 > 0, \quad M_\chi, M_\ell \gg v_\phi, \Lambda_c. \quad (51)$$

For example, $V_\Sigma = 1, m^2 = 1, \kappa = -1, \rho = 0$ gives $M_\chi^2 = 2$ and $M_\phi^2 = -1$ before quartic shifts. This establishes a nonempty tree-level parameter region.

The charge-two condensate has a $\mathbb{Z}_2$ stabilizer in the covering $SU(2) \times U(1)$, but this stabilizer is exactly the identified colour centre. Indeed $[U, -1] = [-U, 1]$ in $U(2)$, so

$$U(2) \xrightarrow{\langle \phi \rangle} SU(2)_c \quad (52)$$

without an additional independent $\mathbb{Z}_2$ gauge factor. The coarse scalar vacuum is still

$$\{\phi^\dagger \phi = v_\phi^2\} / U(1)_g \simeq \mathbb{CP}^1. \quad (53)$$

For nonminimal scalar charge, our symbol Vol_{S^2} denotes the curvature form induced from the covering ($U(1)$) orbit and is normalized by

$$\int_{S^2} \text{Vol}_{S^2} = X_\phi, \quad \omega_2 := \frac{\text{Vol}_{S^2}}{X_\phi}, \quad \int_{S^2} \omega_2 = 1. \quad (54)$$

Consequently, in the covering-group normalization,

$$S_{\text{mix}} = 2\pi\hbar(n_c X_q) \int \omega_3 \wedge \frac{\text{Vol}_{S^2}}{X_\phi} \quad (55)$$

would have coefficient $n_c X_q = 2$ for $n_c = 2, X_q = 1, X_\phi = 2$ [4]. Because the actual faithful subgroup is $U(2) = [SU(2) \times U(1)]/\mathbb{Z}_2$, applying this covering- $U(1)$ normalization to the quotient theory also requires a global match of principal bundles and allowed fluxes. We have not performed that bundle match, so Eq. (55) is a conditional normalization result, not an all-bundle theorem for faithful $U(2)$. This topological statement must not be confused with the local dressing representation. Equation (46) now gives

$$qq\phi^\dagger : \quad 2(1) - 1(2) = 0. \quad (56)$$

Only one global $SU(2)_\phi$ doublet is used. The operator is a doublet, not $\text{Sym}^2 \mathbf{2} = \mathbf{3}$, and no exactly-two microscopic derivation of the Route-E triplet survives.

Fail-closed gate 7.2 (Faithful- $U(2)$ bundle normalization). The missing step is to prove that the determinant line and its first Chern class remain the stated integral generator when the $U(1)$ flux is correlated with the $SU(2)$ -centre 't Hooft flux by the quotient. Until that calculation is supplied, `u2_quotient_global_bundle_normalization_proven=false`.

Fail-closed gate 7.3 (Status of the charge-two branch). The representation, anomaly, and tree-level scalar mass algebra are **conditional but explicit**. The branch is not equivalent to the original charge-one/exactly-two model. It may UV-complete the normalized mixed topological coefficient only if the faithful- $U(2)$ bundle gate closes, while it already fails the specific triplet mechanism; these are logically distinct claims.

8 Running, monopoles, and the all-scale boundary

With two Dirac **3** fermions, one real adjoint scalar, and two complex $\bar{\mathbf{3}}$ scalars, the one-loop deep-UV coefficient is

$$\begin{aligned} b_0^{SU(3)} &= \frac{11}{3}C_2(\mathbf{8}) - \frac{4}{3}N_f T(\mathbf{3}) - \frac{1}{6}T(\mathbf{8}) - \frac{1}{3}(2)T(\bar{\mathbf{3}}) \\ &= 11 - \frac{4}{3} - \frac{1}{2} - \frac{1}{3} = \frac{53}{6} > 0. \end{aligned} \tag{57}$$

Thus this field list removes the abelian Landau pole and remains asymptotically free at one loop. This does not close the generalized symmetry at finite V_Σ . The breaking $SU(3) \rightarrow U(2)$ produces finite-mass nonabelian monopoles. They violate the abelian Bianchi identity and hence the continuous magnetic one-form symmetry is exact only in the decoupling limit where those monopoles are removed. At finite scale it is emergent, as is the separate quark chiral symmetry after the lepton mass splitting. This is consistent with the no-go theorem but prevents an all-scale exact two-group claim.

The remaining nonperturbative requirements are independent:

1. select a mesonic rather than diquark/Pauli–Gürsey vacuum;
2. prove a stable $B = 1$ soliton after the charge-two deformation;
3. compute the monopole-induced violation of winding conservation;
4. evaluate the two UV APS/Dai–Freed torsion characters;
5. produce either a protected lepton-decoupling symmetry or a complete threshold-retuning prescription;
6. recover an exactly-two triplet by a different all-scale group or accept that the charge-two branch addresses only the mixed topological term.

9 Deterministic algebraic and numerical regression

The companion program

```
route_f/code/verify_ap_e3_nonextendible_wzw_su3_uv.py
```

checks, without fitting parameters:

1. the full and reduced stable-splitting bordism groups;
2. Gaussian-quadrature normalization of ω_3 , ω_2 , and their product;
3. the local gerbe derivative, overlap period, Hopf transition, and extension phases;

4. branching of $\mathbf{3}, \bar{\mathbf{3}}, \mathbf{6}, \bar{\mathbf{6}}, \mathbf{8}, \mathbf{10}$ from explicit weights;
5. the quotient parity and exactly-two Diophantine no-go;
6. all dynamical gauge anomalies, the colour Witten parity, and $\widehat{\kappa}_q + \widehat{\kappa}_\ell = 0$;
7. a non-diagonal two-flavour benchmark for $M_q = 0, M_\ell = -3V_\Sigma y$, the scalar split, and $b_0^{SU(3)} = 53/6$;
8. fail-closed booleans for torsion selection, radiative protection, finite-scale one-form exactness, Route equivalence, and physics promotion.

A green mechanical run is compatible with `full_uv_closed=false`. This is deliberate: numerical verification can confirm the displayed algebra but cannot supply missing strong dynamics or an unspecified fermion determinant.

10 Claim ledger and next decisive calculations

Claim	Status	Boundary
Degree-five ordinary differential character on every four-cycle	proved	unique because $H^4(X; \mathbb{R}/\mathbb{Z}) = 0$
$\Omega_4^{\text{Spin}}(S^3 \times S^2) = \mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$	proved	stable splitting; free gravitational factor separated
Two transverse-preimage torsion invariants	proved	$\nu_3 \in \Omega_1^{\text{Spin}}, \nu_2 = \text{Arf} \in \Omega_2^{\text{Spin}}$
UV values of (ϵ_3, ϵ_2)	open	$\epsilon_3 = 0$ follows only with the standard two-colour determinant; ϵ_2 uncomputed
No $SU(3)$ -derived colour-singlet charge-one scalar	proved	quotient parity $2j + q = 0 \pmod{2}$
No exactly-two dressing in this $SU(3)$ embedding	proved	$2X_q = 2X_\phi$ contradicts odd/even parity
Charge-two $\bar{\mathbf{3}}$ scalar mass split	conditional	nonempty tree-level window; thresholds and vacuum require control
Dynamical gauge-anomaly cancellation	proved	vectorlike $SU(3)$ and explicit branched ledger
Light quarks plus heavy leptons	conditional	exact tree identity, no chiral protection
Normalized mixed level $n = 2$ in charge-two branch	conditional	covering-group result; faithful- $U(2)$ bundle/flux matching and local triplet remain open
Route-E-equivalent all-scale UV completion	open	monopoles, strong phase, torsion, radiative decoupling, and triplet mismatch

The ordered next calculations are:

1. build the actual Euclidean UV Dirac/Yukawa operator on the two generator backgrounds in Eq. (27) and compute its APS spectral flow or Dai–Freed eta phase;

2. perform one-loop matching of Eqs. (40) and (49), including all symmetry-allowed counterterms;
3. quantify finite-mass monopole events and the lifetime of the emergent winding charge;
4. decide explicitly whether the charge-two branch is retained only as a mixed-WZW UV candidate or replaced by a different group capable of the exactly-two triplet.

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